The Wrong Unit of Uncertainty: Simultaneous Inference for Repeated Cross-National Surveys

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Abstract

Repeated cross-national surveys are used to make claims about whole trajectories of national response distributions, while uncertainty is computed for single wave contrasts; and when country trajectories calibrate cross-country prediction, the calibration objects are themselves estimates from complex samples. We develop simultaneous inference that attaches uncertainty to both units. Within a country, one design-based band certifies a partially ordered family of trajectory claims — from a single wave pair to persistent decline across a preregistered core of the distribution — with no further correction. Across countries, a clustered conformal band gives finite-sample coverage for survey-estimated trajectories; we prove that correcting toward the latent trajectory is unidentified without design-noise information, and characterize when the correction is unnecessary, not reliably learnable at the available K, or feasible: infeasible across countries, active one unit down. In the European Social Survey a marginal reading flags twenty of thirty countries; six certify net decline over 2018–2024, and over 2002–2024 none certifies a persistent slide — a claim only crisis-scale change reliably clears — while closed testing certifies at least six true span decliners among thirty-three. The same discipline cuts the certified WVS deconsolidation set severalfold, toward post-communist and Arab-Spring states.

1 Introduction

Comparative politics increasingly treats a national public as a *distribution* rather than a mean. Trust in parliament in a country and year is a cumulative distribution function over a response scale, and repeated cross-national surveys—the European Social Survey (ESS; Kaminska and Lynn, 2017), the World Values Survey (WVS), the AmericasBarometer (Cohen et al., 2021)—generate claims about how those distributions evolve across waves. A second use follows: *transporting* the trajectories, predicting one country's from the others', for which conformal prediction promises finite-sample, distribution-free coverage (Vovk et al., 2005; Lei et al., 2018).

Two wrong units. One principle organizes this paper: *the stochastic unit of a guarantee must coincide with the unit of the scientific claim it certifies.* Claims from repeated surveys violate it twice. Uncertainty is routinely reported for a local contrast — one wave pair, one threshold — while the *claim* it is taken to support, "democratic trust is declining," concerns a whole trajectory. (The deconsolidation debate (Foa and Mounk, 2016, 2017; Valgarðsson et al., 2025) is the substantive case we re-examine, not a methodological culprit: its own evidence is cohort comparisons.) The second, subtler wrong unit is the calibration: the objects conformal transport calibrates on are themselves *estimated* from complex probability samples, and existing constructions plug them in as if exact (Dunn et al., 2023; Lee et al., 2026).

The two errors are distinct. Writing $F_{c,r}(t)$ for country c's latent CDF at threshold t in wave r and $\widetilde{F}_{c,r}(t)$ for its complex-survey estimate,

$$\widetilde{F}_{c,r}(t) - \mu_r(t) \;=\; \underbrace{\big(F_{c,r}(t) - \mu_r(t)\big)}_{\text{transport error}} \;+\; \underbrace{\big(\widetilde{F}_{c,r}(t) - F_{c,r}(t)\big)}_{\text{design error}}, \tag{1}$$

where μ_r is the transport center. A band calibrated on the observed $\widetilde{F}$ scores treats the sum as the transport error alone: its spread conflates cross-population variation with survey-design variation that is irrelevant to the latent target, so it can be inefficient (under a scale family, by the familiar factor $\sqrt{1-\rho^2}$), yet any correction subtracting the survey noise can make it too narrow, being itself estimated from few countries.

Contributions. We make four. *First, a procedure for the claim unit*: a design-based simultaneous band over a country's whole trajectory, from which a partially ordered family of claims — from a single wave pair up to persistent decline over a preregistered core of the distribution — is read off one coverage event (Section 3.1). *Second, an impossibility theorem for the calibration unit.* Without information on the design-noise law the latent transport-error law is non-identified, and no band built on the observed curves can uniformly improve on the plug-in conformal band beyond its discreteness slack while remaining valid (Theorem 1). Estimated calibration is not a nuisance to correct away; it is an inferential boundary. *Third, the feasibility boundary of the correction, and a selector that respects it.* Even with the design-noise information supplied, the correction's reliability diagnostic obeys an algorithmic floor $D \geq \sqrt{2/(K-1)}$, so the deployed correction requires $K \geq 94$ exchangeable populations — which the two largest surveys fail in opposite ways, while one unit down (regions within countries, $K \approx 230$–290) the same frozen selector activates (Section 6). The anchor band is finite-sample valid at every K, exact under its symmetric centers and carried into the released leave-one-out deployment by an explicit $K/(K-1)$ inflation, and the selector preserves each branch's guarantee (Theorems 3 and 5). (The clustered band is machinery shared with a companion paper on a non-survey domain, Park, 2026; new here is what the survey layer forces — the impossibility, the floor, the safe selector, and the claim-family deployment.) *Fourth, two reanalyses where the procedures change the substantive picture* (Section 7). For European trust the family certifies net decline in six of thirty countries (2018–2024) against the twenty a marginal reading flags; over the full 2002–2024 record no country certifies persistence, while twenty-three of thirty-three certify both declines and recoveries and eight certify span erosion, all read off one band per country. Globally, applying a trajectory-persistence criterion to the Foa–Mounk "deconsolidation" battery on the WVS trend file (1981–2022) cuts the wave-pair certified set by 2.6–6.5×, and the surviving thirteen-country *certified core* concentrates in post-communist and Arab-Spring-aftermath states.

2 Setup and the two targets

Objects. We observe K source countries, indexed $c = 1, \dots, K$, treated as exchangeable units. Each country has a latent *response-distribution trajectory* $F_{c,r}(t)$, a CDF over a fixed grid of thresholds $t \in \{t_1, \dots, t_T\}$ and survey rounds r. We say *response distribution* deliberately: reading a certified shift as an *attitude* shift additionally requires measurement invariance across waves (Section 8), though every claim below survives monotone recoding of the ordinal scale. We do not see $F_{c,r}$; we see a complex-survey estimate $\widetilde{F}_{c,r}(t) = F_{c,r}(t) + S_{c,r}(t)$, where $S_{c,r}$ is design (sampling) error with a design variance $v_{c,r}(t)^2$ that a stratified primary-sampling-unit (PSU) / replicate-weight bootstrap can estimate. Three curves carry the paper: the latent deviation from the transport center, $G_c = \{F_{c,r}(t) - \mu_r(t)\}_{r,t}$; the design error S_c; and their observable sum $\widetilde{G}_c = \{\widetilde{F}_{c,r}(t) - \mu_r(t)\}_{r,t} = G_c + S_c$, the decomposition (1). The transport score of a country is its worst-case standardized deviation over the trajectory,

$$R_c \;=\; \max_{r,t} \frac{|G_{c,r}(t)|}{\sigma(t)}, \qquad \widetilde{R}_c \;=\; \max_{r,t} \frac{|\widetilde{G}_{c,r}(t)|}{\sigma(t)},$$

with $\sigma(t)$ a fixed modulation profile (the deployed anchor takes $\sigma \equiv 1$, the unstudentized score); R_c is latent, $\widetilde{R}_c$ observed, and $\xi_c := \widetilde{R}_c - R_c$ is the score contamination — a difference of maxima, bounded by $|\xi_c| \leq \max_{r,t} |S_{c,r}(t)|/\sigma(t)$ but not additive noise, and nothing below treats it as centred. Design noise is measured where the additive model holds exactly, at the curve coordinate: with $s^2_{\text{plug}}(t)$ the between-country variance of $\widetilde{G}_c(t)$, $s^2_R(t)$ that of $G_c(t)$, and $\overline{v^2}(t)$ the mean design variance, $s^2_{\text{plug}}(t) = s^2_R(t) + \overline{v^2}(t)$ for design error independent of the latent curve, and the *design share*

$$\rho(t)^2 \;=\; \frac{\overline{v^2}(t)}{s^2_{\text{plug}}(t)} \;=\; \frac{\overline{v^2}(t)}{s^2_R(t) + \overline{v^2}(t)} \;\in\; [0,1)$$

is the governing quantity: $\rho \to 0$ is the design-noise-negligible regime, $\rho \to 1$ the noise-dominated one. The scalar $\widehat{\rho}$ we report and gate on is a conservative aggregate of this profile (the ratio of the t-averaged lower confidence bound on $\overline{v^2}(t)$ to the t-averaged upper bound on $s^2_{\text{plug}}(t)$), a diagnostic rather than a sup-score variance decomposition.

Target and validity. We want a simultaneous band covering a new country's *latent* trajectory (target T2), $\Pr\big(F_{\text{new},r}(t) \in \widehat{B}(r,t)\ \forall r,t\big) \geq 1-\alpha$; the *survey-estimate* trajectory $\widetilde{F}_{\text{new}}$ is target T1. Ordinary clustered conformal on the observed scores is exact for T1 under exchangeability and a symmetric center (Theorem 3); its latent coverage differs by an explicit gap vanishing with the design-noise scale (Theorem 2(b); symmetry alone is not enough), its spread carries design variation — under a scale family, a width inflation by $1/\sqrt{1-\rho(t)^2}$ — and removing that is a deconvolution. Table 1 collects the paper's procedure–target–guarantee triples.

What country exchangeability assumes. The transport band's identifying assumption is a superpopulation postulate — participating countries' trajectory laws are symmetric draws, over a set that is not sampled and enters rounds non-randomly — tempered three ways: the headline certifications of Section 7 do not use it; under approximate exchangeability the Barber et al. (2023) bound controls the gap; and a leave-one-region-out audit finds near-nominal held-out coverage except for the post-communist East (Supplementary

Procedure	Target	Key assumption	Guarantee
Clustered conformal band (anchor)	survey-estimated trajectory	country exchangeability	finite-sample, every K, under symmetric centers (Thm 3)
Same band, latent reading	latent trajectory	+ design-noise tail and anti-concentration	$1-\alpha-$gap, gap shrinking with the design-noise scale (Thm 2b)
Design-aware correction	latent trajectory	design law (A1)–(A3), $K \geq 94$	$1-\alpha-\varepsilon_{K,B}$; deployed level a frozen calibrated bound (Thm 4)
Certification band (§3.1)	within-country contrast surface	survey-design bootstrap	asymptotic simultaneous $1-\alpha$

Table 1: Which procedure guarantees what. The conservative envelope (omitted) shares the anchor's row, containing it by construction; the theory's finite-sample language does not transfer to the certification band.

Material).

The impossibility. The deconvolution is not identified from the contaminated curves alone. The result is stated at the curve level of (1), where the additive model *is* the survey sampling structure; the design noise may be arbitrarily heteroskedastic across thresholds.

Theorem 1 (Non-identification and necessity of design information)**.** *Let the observed deviation curves be $\widetilde{G}_c = G_c + S_c$ as in (1), the latent curves exchangeable with law $\mathcal{L}_G$, the design-noise curves independent of them with law $\mathcal{S}$ in an admissible class that contains $S \equiv 0$ and is closed under one further nondegenerate symmetric convolution. (i) $\mathcal{L}_G$ is not identified from the observed law $\mathcal{L}_G * \mathcal{S}$. (ii) Any band whose radius is measurable in the observed curves alone and is valid for the latent target over the whole admissible class cannot uniformly improve on the plug-in conformal band: on the member $S \equiv 0$, which it cannot distinguish from the contaminated configurations, the plug-in radius is already the tightest valid one up to the conformal discreteness slack, which randomized smoothing exhausts. Uniform improvement requires shrinking the class of noise laws.*

The reading is constructive (proof in the Supplementary Material, Section S1): a band both valid and narrower than the plug-in must supply the design-noise law — exactly what

Section 3's design bootstrap does.

Remark 1 (Beyond surveys). Nothing here is survey-specific: the result holds whenever conformal calibration uses estimated objects — model-based small-area outputs, learned or imputed features, meta-analytic effects. Design or measurement information is the identifying ingredient, its finite-K reliability itself bounded (Proposition 2); confirmed on a non-survey small-area calibration (Supplementary Material).

Relation to existing work. Conformal prediction has met complex surveys for unit-level prediction (Wieczorek, 2023), and cluster-as-unit constructions treat each cluster's data as cleanly sampled (Dunn et al., 2023; Lee et al., 2026); our calibration objects are instead estimated under complex designs (Binder, 1983; Lumley, 2010). Noise-aware conformal methods correct under a *known* noise model (Einbinder et al., 2024; Sesia et al., 2025; Singer et al., 2025); Theorem 1 says that premise fails for design noise — a deconvolution problem (Fan, 1991) — and Proposition 2 that even a supplied law faces a finite-K floor.

3 Simultaneous inference for both units

The two wrong units get two procedures: within-country certification of a whole claim family, and cross-country transport with a correction that activates only where it is provably reliable.

3.1 Within-country certification: one band, a family of claims

The claims of interest differ in their unit. Write $\theta_{a,b}(t) = F_b(t) - F_a(t)$ for the contrast between observed rounds $a < b$ at threshold t — positive where response mass has moved below t, a decline — and $\mathcal{T}_0$ for the preregistered core of thresholds. Table 2 defines each claim as an event for this contrast surface. They form a partial order, not a chain: persistent implies both net (by telescoping) and any-pair, but net and any-pair are incomparable — a dip that recovers certifies an adjacent pair with no net decline, and a slow accumulation

Claim	Event A_j for the surface θ	Certified when
Pairwise $(a, a{+}1)$	$\theta_{a,a+1}(t) > 0 \;\; \forall t \in \mathcal{T}_0$	$\min_{t\in\mathcal{T}_0} \big[\widehat{\theta}_{a,a+1}(t) - c\,\widehat{s}_{a,a+1}(t)\big] > 0$
Any-pair	$\exists\, a :\; \theta_{a,a+1}(t) > 0 \;\; \forall t \in \mathcal{T}_0$	some adjacent pair certifies
Net (span)	$\theta_{1,R}(t) > 0 \;\; \forall t \in \mathcal{T}_0$	the first-to-last contrast certifies
Persistent	$\theta_{a,a+1}(t) > 0 \;\; \forall t \in \mathcal{T}_0,\; \forall a < R$	every adjacent pair certifies
Recovery (a, b)	$\theta_{a,b}(t) < 0 \;\; \forall t \in \mathcal{T}_0$	the reverse inequality, same c
Across countries	conjunction over countries	Bonferroni, or closed testing (§7)

Table 2: The claim family as events for one country's contrast surface $\theta = \{\theta_{a,b}(t)\}$ over R observed rounds. $\widehat{\theta}$ and $\widehat{s}$ are the design-bootstrap estimate and standard error of each contrast and c the single sup-t critical value shared by every entry and both directions, so every certification is read off the one event $\theta \in \widehat{B}$. Persistent implies net and any-pair; net and any-pair are incomparable.

can certify a long span while no single adjacent pair clears the critical value. The claims are not separate tests: each claim, every sub-span, and the reverse direction is a function of one object, the country's surface of wave-pair contrasts, so all are certified on one coverage event.

Proposition 1 (Claim-family transfer)**.** *Let $\widehat{B}$ be a simultaneous $1-\alpha$ band for a country's contrast surface $\theta = \{\theta_{a,b}(t)\}$ over every ordered pair of observed rounds, and let $\{A_j\}$ be any family of claims, each a set of surfaces. Define the certified subfamily $J = \{j : \widehat{B} \subseteq A_j\}$. Then $\Pr\big(\theta \in A_j$ for every $j \in J\big) \geq 1-\alpha$, with no dependence on $|J|$: on $\{\theta \in \widehat{B}\}$, set inclusion delivers every certified claim at once.*

The proof is the line just given; reading each deterministic consequence off one simultaneous region is the post-hoc logic of Scheffé (1953), closed testing (Marcus et al., 1976), and the selection-free bounds of Goeman and Solari (2011); Katsevich and Ramdas (2020). The contribution is its deployment: trend claims in this literature are made claim by claim, and one studentized sup-t band over the whole contrast family removes the corrections a per-claim construction needs — no Bonferroni across a country's pairs, no separate budget for certified recoveries, no endpoint-selection cost — at the price of a wider critical value, whose cost we report (Section 7). The persistent object is covered directly by one functional band (Diquigiovanni et al., 2022) — an estimand distinction, not an error-rate one (Gelman et al., 2012; Benjamini and Hochberg, 1995). The band itself is a design-based sup-t band on wave-pair CDF differences from the stratified-PSU bootstrap (Rao and Wu, 1988; Wolter, 2007), an asymptotic guarantee to which the theory sections' finite-sample language does not transfer.

3.2 Cross-country transport and the safe adaptive band

Theorem 1 says the design bootstrap is the identifying input; the transport method turns it into a band by choosing, target-blind, among three constructions.

Three branches. For calibration curves $\{\widetilde{G}_c\}_{c=1}^K$ with per-country design SD $\widehat{v}_c(t)$ from the stratified-PSU bootstrap: *(i) the clustered conformal band* takes the conformal quantile of the *unstudentized* scores $\max_t |\widetilde{G}_c(t)|$, exact for the survey-estimate target at any K under Theorem 3's symmetric centers; *(ii) deconvolution* studentizes by the finite-K-guarded target scale $\widehat{s}_T^2(t) = \max\big(s^2_{\text{plug}}(t) - [\,\overline{\widehat{v}^2}(t) - z\ \text{SE}\,]_+,\ \text{floor}\big)$, which subtracts a *lower* confidence bound on the design variance ($z = 1.645$ deployed) so the correction cannot over-shrink, and whose width scales as $\sqrt{1-\rho(t)^2}$ coordinate by coordinate under (A1); *(iii) the conservative band* inflates each score by the design SD, $\max_t(|\widetilde{G}_c(t)| + z\,\widehat{v}_c(t))$, which dominates the clustered score pointwise, so the conservative band *contains* the clustered band by construction.

Safe selector (Algorithm 1). A naive selector asks *can we deconvolve* and *do we need to*; the safe selector adds *can we do it safely at this K?* Deconvolution is used only when four target-blind gates pass, all computed from the calibration set: *need* ($\widehat{\rho}_{\text{LCB}} > \rho_0$, the aggregate design share against a frozen operational cutoff), *reliability* ($D = \max_t \text{SE}(\widehat{s}_T^2)/\widehat{s}_T^2 \le \tau_D$, a frozen bound on the finite-K remainder, Proposition 2), *efficiency* (a certified width gain over the conservative band), and *stability* (the guarded scale stays above its floor); otherwise the band is an anchor. All constants are frozen before validation (Supplementary Material), and Theorem 5 shows validity survives any selection between the anchors, the deconvolution branch being charged its own budget only where its gate can open at all ($K \ge 94$). Three

Algorithm 1: Nominal-safe adaptive design-aware band

Input: calibration curves $\{\widetilde{G}_c\}$ (unit = country), design replicates $\{\widehat{v}_c\}$, transport center $\widehat{\mu}$ (leave-one-out mean in the unsurveyed-target default), level α; frozen gate constants and the K-only budget rule $\alpha_{\text{dec}}(K)$

set $(\alpha_{\text{anchor}}, \alpha_{\text{dec}})$ by the frozen budget rule (full α to the anchors when gate B is infeasible at this K);

compute diagnostics s, $\widehat{s}_T$, $\widehat{\rho}_{\text{LCB}}, D, \widehat{\delta}_{\text{UCB}}(D)$; the nested unstudentized anchor radii $r_{\text{clu}}, r_{\text{con}}$ at level α_{anchor}; and r_{dec} at level α_{dec};

if $\widehat{\rho}_{LCB} \le \rho_0$ **then**

  branch $\leftarrow$ clustered;

else if $\widehat{\delta}_{UCB}(D) \le \delta_{\max}$ ***and*** *stable* ***and*** $\widehat{Gain} - \Delta_g > g_{\min}$ **then**

  branch $\leftarrow$ deconvolution;

else

  branch $\leftarrow$ conservative;

if branch $\neq$ deconvolution **and** the center is leave-one-out (the unsurveyed-target default) **then**

  $r_{\text{branch}} \leftarrow \frac{K}{K-1} r_{\text{branch}}$; `// exactness for the deployed center (Section 4)`

return $\widehat{\mu} \pm r_{\text{branch}}$, branch, and diagnostics;

levels must not be confused: the *nominal* $1-\alpha$; the *theorem* level $1-\alpha-\varepsilon_{K,B}$ of Theorem 4, whose remainder is unobservable; and the *validated operational* level the procedure reports when it deconvolves, $1-\alpha-\widehat{\delta}_{\text{UCB}}(D)$, a bound frozen before validation and checked by it. The method ships as one call, `dapcb(cal_errors, v_cal, center, alpha)`, returning band, branch, level, and the target the level refers to.

4 Theory

Theorem 3 anchors the validity architecture and Theorem 5 carries the guarantee through the selector; Theorems 2 and 4 price what the deployed band does not get for free — the latent rather than estimated trajectory, an estimated rather than known noise law. Proofs are in the Supplementary Material, Section S1; each theorem is paired with an automated contract test checking its computational implications.

Theorem 2 (Oracle validity)**.** *Suppose the countries are exchangeable and the design law is* *known. (a) The clustered band is finite-sample valid for the survey-estimate target,* $\Pr(\widetilde{F}_{new} \in \widehat{B}) \geq 1-\alpha$*, using exchangeability only. (b) For the latent target the same band satisfies, for every* $u > 0$, $\Pr(F_{new} \in \widehat{B}) \geq 1-\alpha-\Pr(\xi < -u) - \Pr(\widetilde{R} \in (\widehat{q}-u, \widehat{q}])$*. (c) Under the scale-family condition (A1), the oracle deconvolution band with the true target scale is finite-sample exact for the latent target, with width a factor* $\sqrt{1-\rho(t)^2}$ *of the clustered band at each coordinate, up to* $o(1)$*; some such shape restriction is necessary (Supplementary Material).*

The gap in (b) is a noise-tail plus a local anti-concentration term. Both vanish with the design-noise scale — $|\xi| \leq \|S\|_\infty$, so the tail term concerns the design error alone, and with σ_ξ a scale for ξ the gap is of order $\sigma_\xi^{2/3}$ under variance-only tails and σ_ξ up to logarithms under sub-Gaussian noise — but not by symmetry (a bimodal counterexample, Supplementary Material), and nothing assumes ξ centred: hence we estimate and report the design share rather than assume it away.

Theorem 3 (Exact finite-K validity of the deployed base band)**.** *Let the base band use the unstudentized cluster score of the* observed *curves,* $\widetilde{R}_c = \max_t |\widetilde{G}_c(t)|$ *(*t *ranging over the stacked round-by-threshold grid; the latent* R_c *of Section 2 is unobserved), with a center satisfying the symmetric-construction condition (the Supplementary Material states the four admissible cases). If the countries are exchangeable, then for every* K *the band* $\widehat{B} = center \pm \widehat{q}$*, with* $\widehat{q}$ *the* $\lceil (1-\alpha)(K+1) \rceil$*-th order statistic of* $\{\widetilde{R}_c\}_{c \leq K}$ *— and* $\widehat{q} = \infty$ *when that rank exceeds* K*, i.e. when* $K < \alpha^{-1}-1$ *— satisfies* $\Pr(\widetilde{F}_{new} \in \widehat{B}) \geq \lceil (1-\alpha)(K+1) \rceil/(K+1) \geq 1-\alpha$*, distribution-free, with equality at the stated rank level when the scores are almost surely distinct. The guarantee holds at every* K*; the band is informative only for* $K \geq \alpha^{-1}-1$*.*

Exactness at every K is bought by excluding two asymmetries — in-sample modulation and grand-mean centers containing their own unit — at a width cost of at most 5% on the near-homoskedastic real curves.

Theorem 4 (Estimated-law validity)**.** *Assume a common scale family indexed by the design variance (A1), anti-concentration of the sup-score law near its upper quantile (A2), and* *bounded curves with an unbiased size-B design bootstrap (A3); the appendix states all three precisely. Then (i) with oracle scales the deconvolution band is finite-sample exact for the latent target at every K, and (ii) with estimated scales it satisfies* $\Pr(F_{new} \in \widehat{B}) \geq 1-\alpha-\varepsilon_{K,B}$ *with* $\varepsilon_{K,B} = O\big(\sqrt{\log(KT)/\min(K,B)}\big)$.

The remainder's direction is fixed by construction: the finite-K-guarded scale $\widehat{s}_T^2$ of Section 3 subtracts a lower confidence bound on the design variance, so outside an α_2-probability event the scale error widens the band. Some shape restriction is not removable (the counterexample above); (A1) is one sufficient choice.

Theorem 5 (Safe-adaptive finite-K validity)**.** *Assume the countries are exchangeable and the anchors use Theorem 3's centers (or the leave-one-out center with its $K/(K-1)$ inflation, Proposition S1); for the second clause assume additionally (A1)–(A3). Deploy Algorithm 1 with the unstudentized clustered and conservative branches, scores* $\max_t |\widetilde{G}_c(t)|$ *and* $\max_t(|\widetilde{G}_c(t)| + z\widehat{v}_c(t))$*, at level* α_{anchor}*, and the deconvolution branch at* α_{dec}*. Then, with no condition on the selection rule,*

$$\Pr\left(\widetilde{F}_{new,r}(t) \in \widehat{B}(r,t)\ \forall r,t\right) \geq 1-\alpha \quad (K < 94),$$

$$\Pr\left(F_{new,r}(t) \in \widehat{B}(r,t)\ \forall r,t\right) \geq 1-\alpha-\varepsilon_{K,B} \quad (K \geq 94),$$

the first clause finite-sample, the second carrying Theorem 4's remainder.

Proof sketch. Below the floor, gate B cannot open (Proposition 2(b) is a deterministic identity of the frozen diagnostic), so the selector only ever chooses between the two anchors. Their scores are ordered pointwise, $|\widetilde{G}_c(t)| + z\widehat{v}_c(t) \geq |\widetilde{G}_c(t)|$, so at a common level the conservative order statistic is at least the clustered one and the bands are nested, $\widehat{B}_{\text{clu}} \subseteq \widehat{B}_{\text{con}}$. A miss of whichever band is selected therefore implies a miss of $\widehat{B}_{\text{clu}}$, which Theorem 3 bounds by α: the first clause, with no conditioning on the selection event (Gupta et al., 2022; Yang and Kuchibhotla, 2024). At $K \geq 94$ the budget splits, $\alpha = \alpha_{\text{anchor}} + \alpha_{\text{dec}}$, by a frozen rule of K alone, and the target changes because the assumptions do: under (A1) each observed calibration score dominates its latent counterpart ($\sqrt{s_R^2 + v_c^2} \geq s_R$ pointwise), so the anchor quantile stochastically dominates the latent-score order statistic and the anchors are valid for the *latent* target at α_{anchor} (anchor-domination lemma, Supplementary Material). The non-nested deconvolution branch is charged its own budget by a union bound, $\Pr(\text{miss}) \leq \Pr(\text{miss}, \widehat{J} \neq \text{dec}) + \Pr(\text{miss}, \widehat{J} = \text{dec}) \leq \alpha_{\text{anchor}} + \alpha_{\text{dec}} + \varepsilon_{K,B}$, the last term from Theorem 4(ii) for the always-deconvolve band; the same-order term its leave-one-out centering adds is absorbed into $\varepsilon_{K,B}$ (Supplementary Material). No term conditions on $\widehat{J}$.

Two scope notes. At $K \geq 94$ the level the procedure reports is a validated operational bound (Section 3), not the theorem level. And the unsurveyed-target deployment uses a leave-one-out center, outside Theorem 3's symmetric constructions, but inflating the radius by $K/(K-1)$ restores finite-sample validity at every K, distribution-free (Proposition S1); the released default ships that inflation at a width cost under 3.5%, the frozen gates deciding on the uninflated radii so the sealed validation carries over a fortiori. (The exact split-fold alternative is infeasible at survey scale: halving the calibration set below $\alpha^{-1}-1$ units makes the conformal radius infinite.)

Efficiency. Under (A1) the low-ρ deconvolution-to-clustered width ratio is $1 - \frac{1}{2}\rho(t)^2 + O(\rho(t)^4)$ at each coordinate, proved; the conservative band contains the clustered one pointwise; and selection cannot inflate miscoverage (Theorem 5). The remaining width results, one of them open, are in the Supplementary Material.

5 Simulation and a design-preserving stress test

Simulation establishes two facts: attaching the band to the wrong unit collapses coverage, and the safe selector meets its guarantees on data it has never seen.

5.1 The wrong unit collapses coverage

On a ground-truth simulation ($K = 30$ countries, $T = 6$ thresholds, trajectory length L rounds, errors correlated across rounds and thresholds) we ask each band for *trajectory* coverage, the whole curve-path at once (Figure 1). All three bands are unstudentized and finite-sample, so only the score's unit varies. A band that is marginally valid at the round level covers the trajectory 81.7% of the time at $L = 2$ and 49.8% at $L = 8$; per threshold the figures are 38.2% and 3.5%. Only the country-trajectory band holds its nominal 90% at every length, because the trajectory is a single exchangeable unit. The obvious rejoinder — correct the wrong-unit bands for multiplicity — fails at survey scale: a Bonferroni-corrected per-round band needs a conformal quantile at level α/L, which exists only for $K \geq L/\alpha - 1$ countries (79 at $L = 8$; 479 for the threshold-level band), so at $K = 30$ its radius is infinite for every $L \geq 4$; where it exists ($K = 100$) it is 4–26% wider than the trajectory band for comparable or higher coverage (90–94% against 89–91%). The question is not whether to correct but which unit carries the simultaneous statement at the K surveys have. The same mechanism drives the certification-count collapse of Section 7. The preregistered simulation is reported as produced; the multiplicity-corrected baselines are an additional robustness benchmark.

Severity: what each claim can detect. At ESS-like design noise, 80% power at the *net* claim needs a per-pair decline of 0.02–0.03 CDF points (realistic secular erosion); the *persistent* claim needs 0.06–0.08, crisis-scale, and its power *falls* as the trajectory lengthens. Injecting known declines into the real ESS pairs puts the thresholds at ≈ 0.033 (net) and ≈ 0.075 (persistent), with realized size 0.001–0.007 against a nominal 0.10: markedly conservative, so non-certification is weak evidence of absence at every claim. The powered claim for secular decline is the net span (Section 7).

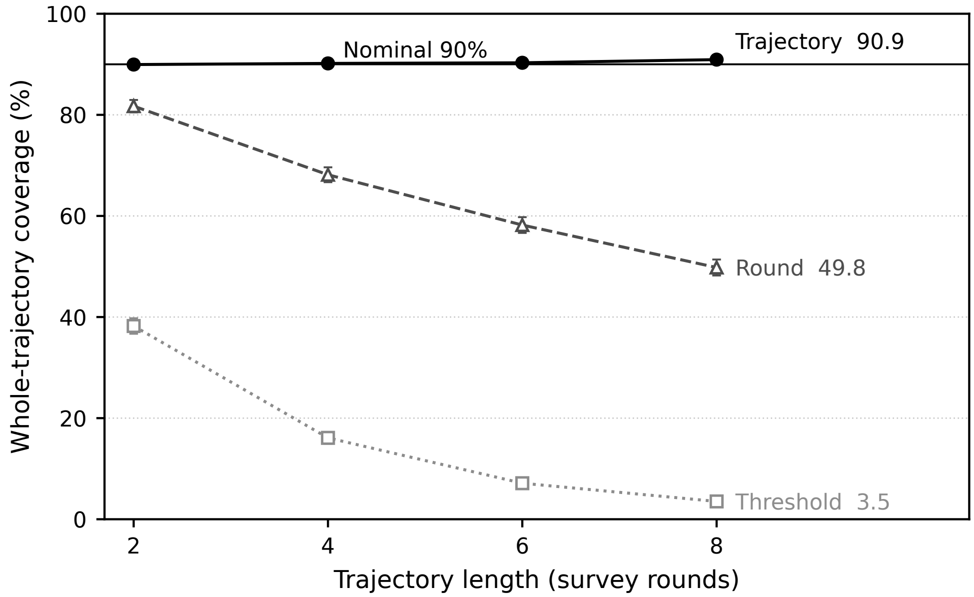

Figure 1: Coverage of a whole country trajectory, by the unit the score is attached to (4,000 replicates, nominal 90%; bars are ±2 Monte Carlo SEs). Only the trajectory-unit band holds its level as the trajectory lengthens.

Validation. The frozen procedure was then evaluated on a sealed set of six unseen data-generating families, script hash recorded before first execution: all 120 cells were compatible with the target coverage (worst 0.878), with zero activations below $K = 94$, as the floor requires. The three-layer validation history — including a first holdout demoted to development after a scorer audit — is in the Supplementary Material.

A design-preserving stress test. Subsampling the real AmericasBarometer, the deployed selector never fires; the $\widehat{\rho}$ turnover at deep subsampling is a sparse-PSU failure of the diagnostic, not a saturation law (Supplementary Material).

6 Evidence from the ESS and the AmericasBarometer

The regime characterization. At the cross-national unit the selector reduces to the clustered conformal band, not by assumption but by its own low-ρ gate: across both surveys and every estimand examined, $\widehat{\rho} \leq 0.22$ on the AmericasBarometer and at most 0.29 in the ESS subgroup scan, well below the frozen need cutoff $\rho_0 = 0.47$. Every $\widehat{\rho}$ we report is a downward-biased *floor* in expectation, the PSU bootstrap resampling m-of-m without Rao–Wu–Yue rescaling (Rao et al., 1992); the rescaled rerun leaves the regime unchanged — maxima 0.28 and 0.22, zero gate activations, every certification count unchanged (Supplementary Material).

Why the deconvolution is unreachable, precisely.

Proposition 2 (Finite-K feasibility of the deployed correction)**.** *The deployed selector activates deconvolution only if a* need *gate* $\widehat{\rho}_{\mathrm{LCB}} > \rho_0$ *and a* reliability *gate* $D \le \tau_D$ *both hold, with* $D = \max_t \widehat{\mathrm{SE}}(\widehat{s}_T^2)/\widehat{s}_T^2$ *and the frozen* $\widehat{\mathrm{SE}}(\widehat{s}_T^2) = \sqrt{2s_{\mathrm{plug}}^4/(K-1) + (\mathrm{SD}_K(\widehat{v}^2)/\sqrt{K})^2}$*. Both gates are bounded by* algorithmic identities of the frozen procedure. *(a)* $\rho(t)^2 = \overline{v^2}(t)/s_{\mathrm{plug}}^2(t) \in [0,1)$ *coordinate-wise, and* $\widehat{\rho}_{\mathrm{LCB}}$ *is its conservative aggregate: the need gate fires only when design noise is an appreciable fraction of the between-population spread,* ρ_0 *being a frozen operational threshold. (b) Since* $\widehat{s}_T \le s_{\mathrm{plug}}$ *by construction,* $D \ge \sqrt{2/(K-1)}$*, so the reliability gate requires* $K \ge 1 + 2/\tau_D^2$*, i.e.* $K \ge 94$ *at the frozen* $\tau_D = 0.147$*. (The Supplementary Material separates the deterministic statement from its kurtosis-dependent interpretation.)*

The two requirements pull apart on real surveys (Figure 2). On the ESS the noisiest subpopulations raise $\widehat{\rho}$ only to 0.29 while $K \le 33$ leaves $D \ge 0.25$: neither gate opens (42 cells, zero activations). The WVS is the mirror image: its full item-coverage sets ($K =$ 95–105) make the reliability gate feasible, the certification samples ($K =$ 59–77) do not, and either way $\widehat{\rho}_{\mathrm{LCB}} \le 0.10$ fails the need gate. The survey large enough to estimate the deconvolution has almost no design noise to remove; the survey with real clustered noise has far too few countries. Nor is the floor an artifact of our thresholds: any *unbiased* variance estimator from K exchangeable populations has relative standard error at least $\sqrt{2/(K-1)}$ at the Gaussian member (Lehmann–Scheffé; Supplementary Material), so any correction demanding relative reliability τ from such an estimate needs $K \ge 1 + 2/\tau^2$ — the frozen

τ_D is one instantiation, 94 its intercept, $\sqrt{2/(K-1)}$ the law for this estimator class. With the proved width gain $1-\sqrt{1-\rho^2}$, the (K,ρ) plane splits into three regimes — correction *unnecessary*, *unlearnable*, *feasible* — and this paper's datasets occupy the three corners.

Where the correction does live: small areas, on this same survey. The barrier is about the unit, and the ESS contains a finer one: a region's departure from its own *national* distribution, $D_g = F_g - F_{\text{nation}}$, with the transport band calibrated across regions and the design SD of the difference from one stratified-PSU bootstrap of the whole country (Supplementary Material). There the design share is not marginal — $\widehat{\rho}_{\text{LCB}}$ runs 0.26–0.52 and clears ρ_0 in eight of twenty-three *unit-rounds* (minimum unit size × round) — and both gates open once $K \gtrsim 230$: units of forty to sixty respondents, rounds 9 and 10, $K = 228$–287. There the *frozen* selector activates the design-aware branch for the first time on real survey data, a feasibility demonstration rather than a validation of latent-target coverage (the latent departure is unobserved): a band 21–27% narrower than the conservative envelope at its validated operational level 0.88, with certified width-gain lower bounds of 0.16–0.22 across twelve independent bootstrap streams.

The regions nest in twenty-seven countries, nine each, so the objection this paper would itself raise is that the effective unit is the country. Holding out whole countries is the diagnostic: calibrating on the other twenty-six countries' regions leaves the plug-in anchor at 0.89–0.90 against its 0.90 target in all twelve cells, within 0.015 of leave-one-region-out calibration. Coverage *conditional* on the country does not transfer (worst held-out countries 0.53–0.63): the guarantee is marginal over regions and presumes their exchangeability, which shared national dependence can violate — the diagnostic bounds that concern, it does not remove it. In the activating cells the selected band covers 0.93–0.98 against its 0.88 on the observed departure, latent-target validity resting on (A1) as everywhere in that branch. Two scope conditions: reaching $K \gtrsim 230$ pools regions coded at different NUTS levels (among the fourteen countries sharing one level the need gate opens but $K \leq 140$ keeps the reliability

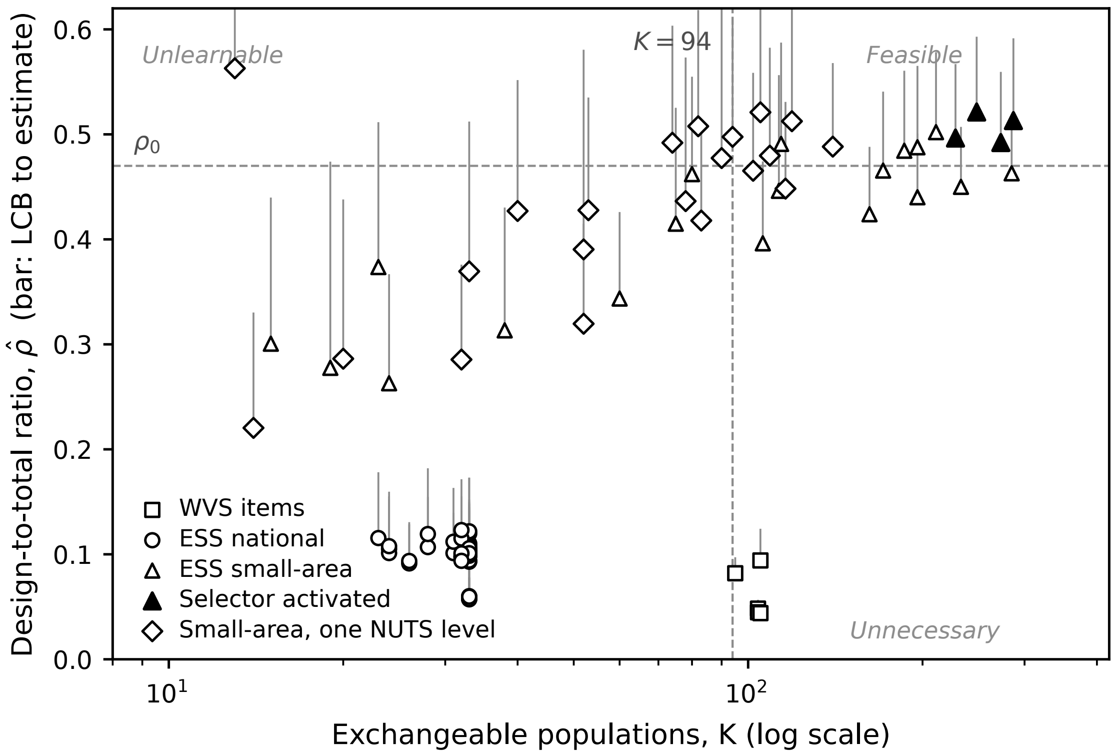

Figure 2: The feasibility frontier in the $(K, \hat{\rho})$ plane. Each marker is a real-data cell at its lower confidence bound, with a bar to the point estimate; filled markers are the four cells where the frozen selector activated. Dashed rules are the frozen boundaries, the need cutoff ρ_0 and the reliability floor $K = 94$.

gate shut), and round 11 never activates (Supplementary Material). The correction is thus reachable in principle and unreachable across countries: it lives at the small-area scale, where Theorem 1's remark said it would.

7 Political reanalysis: how much decline survives simultaneous inference?

We analyze trust in parliament (`trstprl`) and satisfaction with democracy (`stfdem`) in ESS rounds 9–11 (2018–2024, the window whose integrated files ship PSU/stratum identifiers), then extend every trajectory to the full record (rounds 1–11, 2002–2024), asking, claim by claim, how many countries survive. The comparative literature has grown skeptical of a continent-wide trust decline (Norris, 2017; Voeten, 2017; Wuttke et al., 2022); we add a formal procedure that says so and a diagnosis of why the over-reading arises: the dominant trend estimates (Valgarðsson et al., 2025) rest on latent-trait models with no design-based uncertainty layer (Tai et al., 2024), and latent-mood estimation (Claassen, 2019, 2020a; Kołczyńska and Bürkner, 2026) pools countries into model-based credible intervals, where our bands give model-free coverage at a stated claim; the two are complements. Certification levels here are design-bootstrap simultaneous levels, asymptotic in the within-country samples (Table 1).

The counts collapse across claims. For `trstprl` over rounds 9–11, a marginal pairwise reading flags 20 of 30 countries; an any-pair design-aware band drops this to 12; net first-to-last decline holds in 6 (Austria, Belgium, Estonia, the United Kingdom, Greece, the Netherlands); and the *persistent, design-aware simultaneous* band over the preregistered low-trust core certifies one, Greece, which also survives Bonferroni across countries. *Greece fielded only rounds 10 and 11 here*, so its trajectory is a single pair and the top claims coincide for it (as for Czechia and Israel, uncertified). Among the 27 two-pair countries none certifies persistence, while the United Kingdom, Belgium, and the Netherlands pass the *plug-in* persistent claim and are demoted by the design-aware layer alone. The headline is therefore two numbers: net decline in six of thirty, and *zero* of the twenty-seven countries observed across multiple pairs certifying persistence. The severity analysis (Section 5) explains why: the persistent claim has 80% power only against crisis-scale declines, so its non-certifications are weak evidence of absence. For `stfdem` the pattern repeats (23 marginal, 14 any-pair, 8 net, one persistent: Greece, again one pair, failing Bonferroni there). The literature's Greek legitimacy collapse (Armingeon and Guthmann, 2014; Torcal, 2014) concerns 2009–2015; the certified pair is 2020/21→2023/24.

The long window: the secular question, asked properly. Extending every trajectory to its full record (rounds 1–11; median six pairs, up to ten; rounds 1–8 weights-only; the sample-design-file merge changes no net or persistent count — Supplementary Material) reframes the twenty-year narrative in both directions. Zero of the thirty-three countries with three or more usable rounds certify persistent decline over the preregistered core across their full record, plug-in included. Reversing the inequality certifies *recoveries*, and across the 1,173 ordered contrasts of the trust series certified movement is directionally balanced (193 declining against 212 rising). Inside the same band, at the same α, twenty-three of thirty-three countries certify at least one decline and at least one recovery on each outcome (twenty on both); the two directions share one critical value, so no within-country correction is needed. The count across countries is a conjunction of thirty-three coverage events and carries no familywise control by itself; the closed-testing bound below supplies it. The certified record therefore argues against describing European trust as a uniformly monotone slide — on the positive evidence of coexisting certified declines and recoveries, not on the persistent rung's non-certifications, which its severity makes weak evidence of absence.

Under Proposition 1—one band, every ordered contrast and both directions at $\alpha = 0.10$ jointly—net erosion over the whole span certifies in eight countries (Cyprus, Spain, the United Kingdom, Greece, Hungary, Israel, Italy, Ukraine; Figure 3, Table 3). Against the per-claim construction the joint statement costs France at the span claim ($9 \to 8$) and six countries at the any-pair claim ($28 \to 22$); the persistent zero holds under both constructions (Supplementary Material).

The across-country count itself can now carry a guarantee. Inverting each joint band over its level yields a per-country certification p-value; closed testing across the thirty-three (Goeman and Solari, 2011) then certifies, with 90% simultaneous confidence, that *at least six* truly declined over their span, on each outcome, the six nameable at no further cost (Supplementary Material). The Simes local tests behind that bound need the countries' p-values independent or positively dependent — a design assumption (independent sampling mechanisms given the finite populations), not a property of separate bootstrap streams; Bonferroni local tests, which assume nothing about dependence, return the same six on

Table 3: Certified net decline in parliamentary trust, 2002–2024, read off each country's joint band (every ordered contrast, both directions, $\alpha = 0.10$). *Decline* $\geq$: simultaneous lower bound over the preregistered core, in CDF points. *Contrasts*: ordered round pairs certified declining, of all admissible; their share is the endpoint-free erosion index. The certified set is stable under weights-only design-effect inflation up to roughly 1.3; above that Spain, the marginal member, drops out (Supplementary Material).

country	rounds	decline $\geq$	contrasts	share
Ukraine	2–11	25.9	7/15	0.47
Cyprus	3–11	6.2	13/21	0.62
United Kingdom	1–11	6.1	11/55	0.20
Greece	1–11	3.1	8/15	0.53
Israel	1–11	2.6	10/28	0.36
Italy	1–11	1.6	5/15	0.33
Hungary	1–11	1.1	9/55	0.16
Spain	1–11	0.6	32/55	0.58

both outcomes, as does the rounds 1–8 design-file upgrade (Supplementary Material). The wrong-unit ladder thus closes at the top: threshold, wave, trajectory, country, count.

Under-detection shows in the ledger of contrasts: Hungary certifies its 22-year span while only 9 of its 55 ordered contrasts certify, and the United Kingdom 11 of 55 — erosion a reading built from adjacent waves would mostly miss. Because every endpoint choice is certified at once, the band also delivers the *share of a country's ordered contrasts that certify a decline*: a lower bound, comparable within a country but not across (the shared critical value increases strongly with record length while the denominator grows quadratically), so we do not rank on it. Share and magnitude differ: Spain certifies 32 of 55 contrasts yet clears its span by only 0.6 CDF points. Ukraine's certified span decline of at least 25 points runs from the post-Orange-Revolution wave of 2005 to a wartime sample: a certified statement about those two waves.

The reclassified countries are the weak-signal, higher-noise cases—*certified by the marginal plug-in but inconclusive under the design-aware band*, never "false positives," a term we reserve for simulation with known truth.

Two confounds this window cannot exclude. Round-10 fieldwork coincided with the pandemic's rally-round-the-flag movements, so "persistent decline through this window" is a claim made against a rally; and nine countries switched to self-completion at round 10, confounding wave-pair shifts with mode effects, which no design bootstrap absorbs. A mode audit from the data's own mode variable bounds it: the Greek pair is mode-constant and net counts are unchanged when the switchers' confounded pairs are excluded (Supplementary Material, with a preregistered age-group analysis that finds no pooling artifact).

A second, global target: Foa–Mounk deconsolidation. We preregistered a reanalysis of the "democratic deconsolidation" thesis (Foa and Mounk, 2016, 2017) on the WVS trend file (1981–2022, 59–77 countries with ≥ 2 waves), recoding the five-item battery so that deconsolidation is a persistent decline over each scale's preregistered core. The harmonized trend file provides no design metadata, so this is an external substantive stress test of the thesis, not a second validation of the method; the ESS carries the design-based validity story. The verdict is calibration, not denial, but the surviving phenomenon is not the one the thesis was about. A *marginal wave-pair reading of the same data*, constructed to instantiate the bottom rung, flags 36–64 countries per item against 6–17 (8–24%) at the persistent rung. That 2.6–6.5× ratio compounds two changes: holding the inference layer fixed, the rung alone accounts for 1.7–4.8× (plug-in) or 1.9–4.8× (design-aware), and propagating design uncertainty at fixed rung accounts for the remainder. Three caveats. The trend file carries no PSU/stratum identifiers, so the weights-only bootstrap omits within-PSU dependence (variance-inflating) and stratification (variance-reducing); where the former dominates, as is usual, certified counts are *inflated*: at variance ×1.5 and ×2.0 the certified core holds at 13 and 12 countries and the gap widens to 3.0–8.3×, so the reported gap is then a lower bound. The per-item lists carry no across-country familywise control, so we aggregate to multi-item co-certification before interpreting geography. And the certification sets sit below the $K \geq 94$ bound, so every WVS band uses the full-α anchors of Theorem 5.

The certified core: where the surviving deconsolidation lives. Aggregating per-item certifications yields a positive characterization (Figure 4): of 38 countries certified on ≥ 1 item, 13 form a certified core on ≥ 2. The core is post-communist and Arab-Spring states, not the mature democracies the thesis concerned, formalizing with per-country error control the decline-from-euphoric-baselines pattern the post-communist literature described qualitatively (Mishler and Rose, 1997; Inglehart and Catterberg, 2003). Three qualifications travel with it. Five core members are electoral autocracies in the V-Dem classification, where falling stated support reflects autocratic legitimation rather than deconsolidation in the consolidated-democracy sense. The West enters the core exactly twice (Finland and Switzerland, on the strongman/army pair, which we flag as a probable administration artifact): no consolidated Western democracy certifies on support for a democratic system, consistent with Wuttke et al. (2022), though not that the West is absent. And the geography is partly window-contingent: recertifying on fieldwork ending by 2017 keeps eight of the thirteen but drops all three Arab-Spring cases, so the post-communist component is window-stable and the Arab-Spring component rests on WVS wave 7. Certified attitudes do not forecast institutions (core membership predicts subsequent V-Dem electoral-democracy decline no better than the marginal flag, Fisher $p = 0.89$, a comparison too small to establish the decoupling recent panel work expects, Ouattara and van der Meer, 2023; van der Meer and Janssen, 2025): certification is descriptive error control, not a forecast (item-level caveats in the Supplementary Material).

8 Discussion

Uncertainty in claims about repeated cross-national surveys is usually computed for the wrong unit — twice. We gave a multiplicity-honest simultaneous band for the claim unit and a hard boundary for the calibration unit: non-identified without the design-noise law (Theorem 1), unreachable for the deployed correction at cross-national scale with it (Proposition 2). At the national unit the method reduces exactly to clustered population conformal — certifying, with the reported ρ, that the simple band is the right one; the deconvolution regime needs many exchangeable units and comparable design noise, never both across countries but reachable one unit down (Section 6).

Limitations. Four scope conditions (Supplementary Material). The machinery prices sampling error only: measurement non-invariance (wording changes, house effects, mode switches) produces exactly the shifts the bands certify, so certifications straddling such changes are conditional on measurement constancy. The deployed m-of-m PSU bootstrap (Rao et al., 1992; Wolter, 2007) biases design variance downward; the Rao–Wu–Yue rescaled rerun leaves every certification count and cross-national gate unchanged, while the small-area activation set is bootstrap-sensitive in the disclosed direction (Supplementary Material). Transport-band exchangeability is a superpopulation postulate over a non-sampled, panel-attriting country set, and holds worse for the post-communist bloc. And without design metadata (the WVS trend file) only weights-only replication is possible.

The general lesson. Attaching uncertainty to the right unit changes substantive conclusions: no persistent decline certified among thirty-three ESS countries over the full record, yet span erosion certified in eight, mostly invisible to adjacent-wave readings; a certified deconsolidation set several-fold smaller, concentrated outside the consolidated West. The pattern is not confined to surveys: wherever prediction is calibrated on estimated objects (Remark 1) the calibration targets carry their own error, and the claim unit is often coarser than the trend asserted. The right response is not to widen every band reflexively but to state the strongest object the data support, correct the estimation noise when it can be identified, and abstain when it cannot.

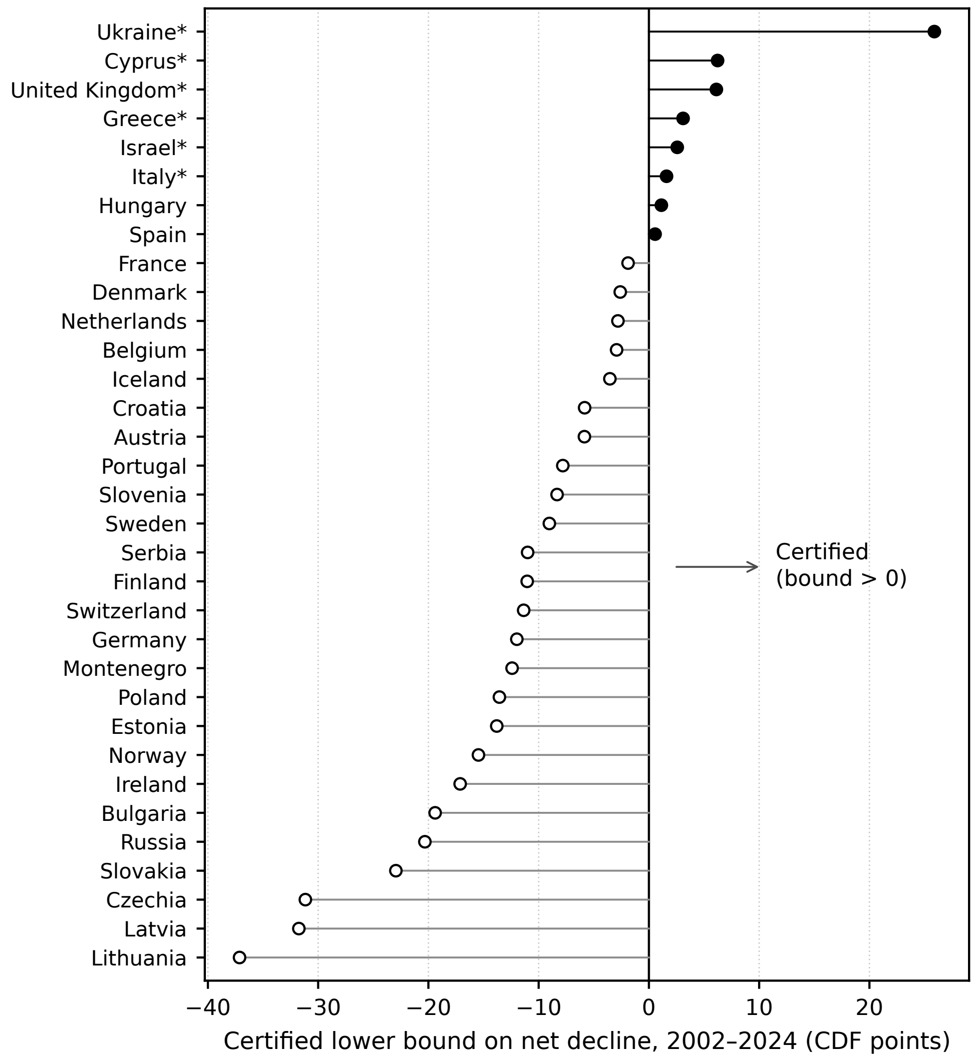

Figure 3: Each country's simultaneous lower bound on the net (first-to-last) decline in parliamentary trust, 2002–2024, read off its joint band at $\alpha = 0.10$. Filled markers: the eight countries whose bound exceeds zero; asterisks: the six named by the closed-testing prevalence bound. Negative bounds say how far a country sits from certifying, not that it improved.

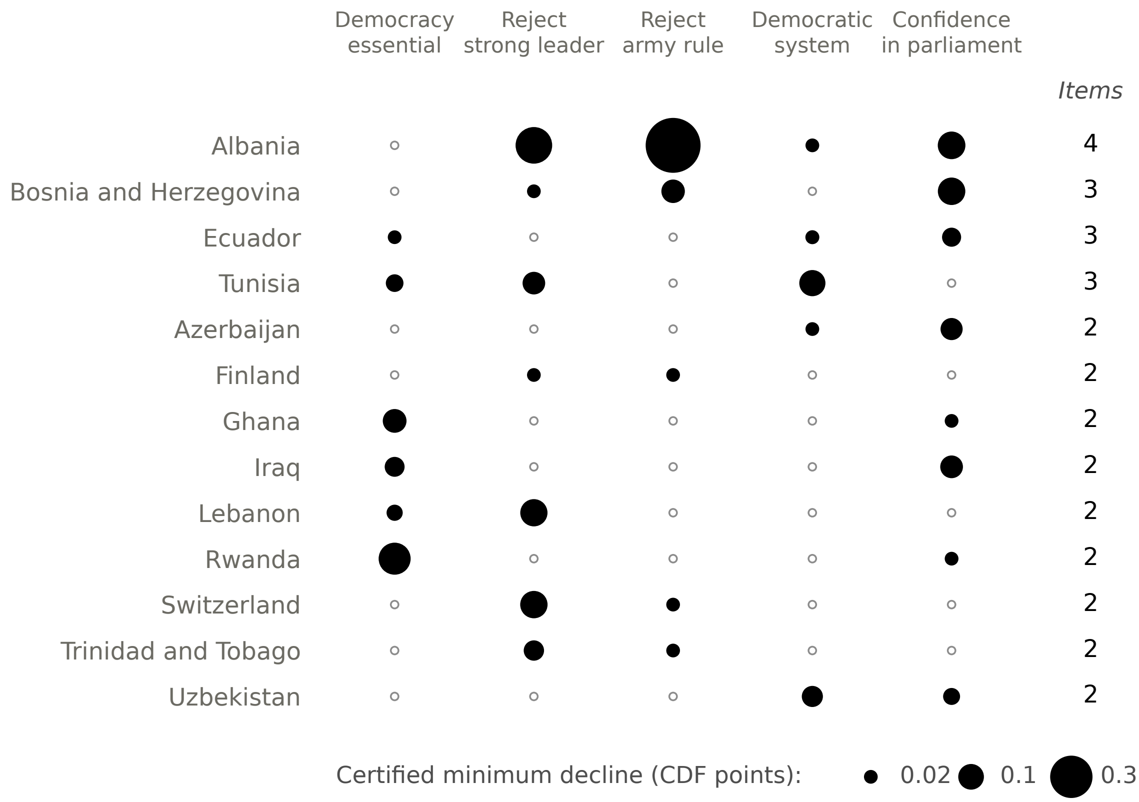

Figure 4: The thirteen-country certified core, by evidence strength (WVS 1981–2022; $K =$ 59–77 per item). Circle area is the persistent band's simultaneous lower bound on the per-pair decline, in CDF points; open circles mark items that do not certify. The full thirty-eight-country matrix is in the Supplementary Material.

Funding

This research received no specific grant from any funding agency, commercial or not-for-profit sector.

Acknowledgments

Generative-AI disclosure (text and proofs): the author used Claude (Anthropic), accessed through the Claude Code command-line interface between July and September 2026, to draft and revise portions of the manuscript text and to help construct, revise, and check proof arguments, working from the author's specifications and from external review feedback. Every mathematical claim and proof was reviewed and verified by the author, and each theorem is additionally paired with an automated contract test in the replication package checking its computational implications. The author takes full responsibility for the content of this article. The corresponding disclosure for code and data analysis appears in the Data Availability Statement.

Data Availability Statement

Replication code for this article is available at `https://github.com/kunhailP/design-aware-conformal`; the package will be deposited in the Political Analysis Dataverse with a version tag upon conditional acceptance. Generative-AI disclosure (code and data analysis): substantial portions of the analysis and replication code, including the figure-generation code, were written with the assistance of Claude (Anthropic; Claude Code command-line interface, July–September 2026), under the author's direction; all AI-assisted code is public in the repository, was reviewed by the author, and is exercised by the package's contract-test suite and its bit-identical reproduction checks. Simulation, theory, and benchmark results are deterministic under fixed seeds with no external data, and each theorem is paired with an automated contract test checking its computational implications. The survey microdata (European Social Survey ERIC, 2024; Haerpfer et al., 2024; LAPOP Lab, 2023) are licensed to their providers and cannot be redistributed; all code that runs on them is public, and the replication package documents the exact file editions, variables, retrieval steps, and sha256 checksums, alongside the public V-Dem and Claassen inputs (V-Dem Institute, 2025; Claassen, 2020b). Three headline analyses have been verified to reproduce the committed result tables bit-identically from the raw licensed files in two independent environments. The editor has been notified of the restricted access at submission.

Competing Interests

The author declares none.

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